

1. Document Overview

This document specifies the communication protocol for the RDXXB series radar and display control system.

2. Communication Agreement

The communication between the radar and display control system uses the UDP network protocol.

The display control system's IP address is: 192.168.1.2, with the local port number 8000;

The radar's IP address is: 192.168.1.3, with the local port number 7000.

For data content exceeding one byte, it is processed in little-endian order, meaning the high byte comes after the low byte.

The display control system broadcasts commands to 255.255.255.255:11111, and the processing machine replies with the network configuration response frame to the corresponding IP and port.

3. Data Protocol

3.1 Frame Format

All communication data frames follow the format outlined below.

Logo	Byte Count	Data Types	Data Content
Frame Header	2	unsigned Short	0x55AA
Identifier	2	unsigned short	0xFF01Tracking target information frame 0xFF02: Search target information frame 0xFF03Status information frame 0xFF04Parameter update frame 0xFF05Scan mode switching command frame 0xFF30Reserved (This content is prohibited from being sent)

			0xFF31Reserved (This content is prohibited from being sent) 0xFF32Reserved (This content is prohibited from being sent) 0xFF33Network configuration command/response frame 0xFF34Reserved (This content is prohibited from being sent)
Frame Count	4	unsigned int	0x00000000~0xffffffffThe loop is synchronized by the receiver.
Frame Content Length	4	unsigned int	The size of the frame content, in bytes.
Frame Content	n		
Protocol Identifier	2	unsigned char	1:RD01 2:RD02 3:RD03 ... 8:RD08 Different radar identifier protocols do not guarantee compatibility. If the software receives a protocol with an invalid identifier... Discussion, simply ignore.
Protocol Major Version Number	1	unsigned char	Main version number:1,2,3... If significant changes occur, the major version number will be upgraded; a higher major version number indicates a more sophisticated protocol. New software is compatible with older protocols, but older software is not guaranteed to be compatible with the new protocol (forward compatibility). (Incompatible with backwards)

			default value:2
Protocol Sub-version Number	1	unsigned char	Sub-version number:0,1,2... If general changes occur, upgrade the sub-version number. If the major version number is the same, upgrade the sub-version number. A larger number indicates a newer protocol, ensuring that new software is compatible with older versions of the protocol, and older software is compatible with theProtocol (forward and backward compatible) default value:13
Checksum	1	unsigned char	Sum all data except the frame header and take the least significant byte. (uint8_tAccumulation)
total	16+n		

3.2Frame content

3.2.1Mode command frame

The mode command frame contains three operating modes: standby, search, and track. This frame is sent to the radar by the display and control system.

The radar defaults to standby mode when powered on.

The radar does not respond to repeated commands.

1) Standby/Search Mode

surface Standby/Search Frame

Logo	byte count	Data Types	Definition
mode type	1	unsigned char	0x00Standby mode 0x11Search mode 0x33:Leveling mode

			(Level the machine before operation, and use it in machine scan or standby mode. Set the servo mode to loop scan mode in the issued command. Do not perform other operations during the leveling process. When a signal is received...) 3.2.2 Status information The mode is 0x00 or 0x33, (Leveling complete)
spare	4	char	for internal use only. 0
Frequency Selection	1	unsigned char	Value range: [0, 20] See appendix for details. 1 Frequency point correspondence table default value: 10
Silent Zone 1 Starting Angle	2	unsigned short	0°~360° precision 0.1°, invalid 7FFFH Range of values [0, 3600] Corresponding status information frame: Radar front end 1 Azimuth Angle like 180.5°, Send value 1805
Silent Zone 1 End Corner	2	unsigned short	0°~360° precision 0.1°, invalid 7FFFH Range of values [0, 3600] Corresponding status information frame: Radar front end 1 Azimuth Angle
Silent Zone 2 Starting Angle	2	unsigned short	0°~360° precision 0.1°, invalid 7FFFH default value: 0x7FFF
Silent Zone 2 End Angle	2	unsigned short	0°~360° precision 0.1°, invalid 7FFFH default value: 0x7FFF
Distance compensation	2	Short	accuracy 0.1 Default fill 0 like 2.3 Meters, sent value: twenty three
Azimuth compensation	2	Short	accuracy 0.1 Default fill 0 like 1.2°, Send value: 12
Pitch Compensation	2	Short	accuracy 0.1 Default fill 0 like 1.1°, Send value: 11
Scan Cycle	2	Short	Both standby and search modes are effective, with high accuracy. 0.1 Default fill 3.2s, default value: 32

Servo Mode	1	unsigned char	Both standby and search modes are effective. 0x11Represents pointing mode 0x33Represents circular scanning mode 0x77Represents pulling out the locking pin 0x88Represents inserting a pin 0xFFRepresents idle mode Other instructions will be executed according to the idle instruction. Note: Standby command0xFF;Search/leveling commands:0x33 Pointing instruction:0x11
Servo Pointing Angle	2	unsigned Short	Both standby and search modes are effective. 0~3600,represent0~360.0degree, accuracy0.1The range of values like180.5°, Send value1805
System Control Flags	1	unsigned char	Only valid in standby mode 0x00This means that the operation corresponding to the system control code will not be executed. 0x01This represents the operation corresponding to the system control code. Others, according to0x00implement
System Control Code	1	unsigned char	Only valid in standby mode 0x01The system control operation is to shut down the radar. Other, no action.
Manual Pitch Enable	1	char	0x00 :Disable0x011:Enable (Values are:)0x011At this time, the angle obtained by leveling will no longer be used, and the set array will be used instead. (Surface mounting pitch angle)

			default value:0
spare	8	char	
longitude	8	double	reserve6Decimal places (setup point)GPSinformation)
latitude	8	double	reserve6Decimal places (setup point)GPSinformation)
high	8	double	reserve6Decimal places (setup point)GPSinformation)
range	1	char	0x11:7.5km;0x22:15km default value:0x11
spare	256	char	
3 Km Inner Height Lower Limit	2	short	accuracy1mUsed for rejection3kmWithin (inclusive)3kmTargets at altitudes lower than this height default value:0
3 Km External Height Lower Limit	2	short	accuracy1mUsed for rejection3kmTargets at altitudes lower than this height default value:100
Height Limit	2	short	accuracy1mRemove targets with a height not less than this value. default value:600
Speed Lower Limit	2	unsigned short	accuracy0.1m/sEliminate targets with absolute speeds lower than this value. default value:0
Speed Limit	2	unsigned short	accuracy0.1m/sEliminate targets with absolute speeds greater than this value. like12.3 m/sSend value:123 default value:600
spare	2	unsigned char	
Target Reporting Distance Limit	2	unsigned short	accuracy1mRemove targets that are more than this distance. default value:7500
Target Reporting Distance to Lower Limit	2	unsigned short	accuracy1mRemove targets that are less than this distance. default value:200

spare	30	unsigned char	
Roll Angle Installation on Array	4	float	Accurate to the decimal point2Position, corresponding to the built-in inclinometerXThe axis, viewed along the radial direction, is lower on the left and lower on the righHigh is positive, and left higher than right is negative. (3.2.2Status information:Roll angle of the formation) default value:0.0
Elevation Angle of Array Installation	4	float	Accurate to the decimal point2Position, corresponding to the built-in inclinometerYThe axis, viewed along the radial direction, is lower in the front and lower in the bacHigh is positive, high in the front and low in the back is negative. (3.2.2Status information:Pitch angle of the array) default value:0.0
North Corner of the School	4	float	Accurate to the decimal point2Bit Value range: [0,360] default value:0.0 InstallGPSAnd if the heading validity mark is valid, the calculation shall be performed with reference to Appendix II.
Autonomous identification enablement	1	char	0: Not enable1Enable (If not enabled, target identification will only be performed by tracking the flight path; otherwise...) Search tracks are also identified, and the search period will be longer. default value:1
Autonomously Identify the Maximum Number of Targets	1	char	scope:1~2 default value:2
reserve	30	char	
total			

2) Tracking Mode

Label	Byte Count	Data Types	Definition
Mode Type	1	unsigned char	0x22: Tracking mode
Tracking Type	1	unsigned char	0x11: Enter tracking 0x22: Cancel tracking
Number of Targets Tracked by Radar	2	unsigned short	Fixed as 1
Spare	128	char	
Target Backup	16	char	
Target Number	2	unsigned short	
Target Distance	4	float	
Target Location	4	float	
Target Pitch	4	float	
Target Height	4	float	
Target Radial Velocity	4	float	

3.2.2Status information frame

Users can selectively display content as needed.

The period for sending this status frame is...200msLeft and right (not a strict time interval).

surface4Status information frame

Label	Byte Count	Data Types	Definition
Work Mode	1	unsigned char	0x00: Standby 0x11: Search 0x22: Track 0x33: Leveling
Command Execution Status	1	unsigned char	0x00: No new commands received 0x11: Executing 0x22: Execution successful 0x33: Execution failed, Other, invalid state
Fault Type	1	unsigned char	0x00: No faults 0x01: Command parsing failed 0x02: Command could not be executed, Other, custom faults
Radar Model	1	unsigned char	0x01–0x10: Corresponding to RD01–RD10
Number of Radar Front-End IDs	1	unsigned char	Fixed value: 0x01
Radar Front-End Network Status	1	unsigned char	0: Not connected 1: Connected
Radar Front-End Scan Cycle	4	unsigned int	
Radar Front-End Azimuth Angle	4	float	[0, 360]
Radar Front-End Pitch Angle	4	float	[0, 360]
Radar Front-End Operating Frequency	2	unsigned short	0–20, corresponds to 2GHz, 20 corresponds to 9.8GHz

Label	Byte Count	Data Types	Definition
Spare	16	char	
Radar Front-End Silent Zone1 Rise	2	short	0°–360° precision 0.1°, invalid 0x7FFF (Valid scope: [0,360]) Such as received value 1805, expressed: 180.5
Radar Front-End Silent Zone1 Knot	2	short	0°–360° precision 0.1°, invalid 0x7FFF (Valid scope: [0,360]) Such as received value 1805, expressed: 180.5
Radar Front-End Silent Zone2 Rise	2	short	0°–360° precision 0.1°, invalid 0x7FFF (Valid scope: [0,360]) Such as received value 1805, expressed: 180.5
Radar Front-End Silent Zone2 Knot	2	short	0°–360° precision 0.1°, invalid 0x7FFF (Valid scope: [0,360]) Such as received value 1805, expressed: 180.5
Fan Status	1	unsigned char	Bit[1:0]: Indicates fan1 state, Bit[3:2]: Indicates fan2 state, Bit[5:4]: Indicates fan3 state. 0 indicates normal, 1 indicates an anomaly.
Frequency Synthesis State	1	unsigned char	Bit0-expressLO1state, Bit1-expressLO2state, 0: Indicates an anomaly, 1: Normal
Sub-array1 Temperature	2	short	accuracy1 Celsius
Sub-array2 Temperature	2	short	accuracy1 Celsius

Label	Byte Count	Data Types	Definition
Sub-array3 Temperature	2	short	accuracy1 Celsius
Sub-array4 Temperature	2	short	accuracy1 Celsius
Sub-array5 Temperature	2	short	accuracy1 Celsius
Sub-array6 Temperature	2	short	accuracy1 Celsius
Sub-array7 Temperature	2	short	accuracy0.1A
Sub-array8 Temperature	2	short	accuracy0.1A
Information Status	1	unsigned char	Bit0-Representation and ChannelADOver-range0-This indicates that the measurement range has not been exceeded.1-This indicates that the range has been exceeded. Bit1-Indicates pitch difference channelADOver-range0-This indicates that the measurement range has not been exceeded.1-This indicates that the range has been exceeded. Bit2-Indicates azimuth difference channelADOver-range0 -This indicates that the

Label	Byte Count	Data Types	Definition
			measurement range has not been exceeded.1-This indicates that the range has been exceeded. Bit3-Indicates the backup channelADOver-range0 -This indicates that the measurement range has not been exceeded.1-This indicates that the range ha Bit4-expressQDRInitialization complete.0 -Indicates incomplete.1-This indicates completion. Bit5- expressDDR-0Initialization complete.0 -Indicates incomplete.1-This indicates completion. Bit6-express DDR-1Initialization complete.0 -Indicates incomplete.1-This indicates completion. Bit7-expressPCIeLink status,0 -Indicates incomplete.1-This indicates completion.
Signal Voltage1	2	unsigned short	accuracy1V
Signal Voltage2	2	unsigned short	accuracy1V
Signal Voltage3	2	unsigned short	accuracy1V
Temperature at Several Locations	4	float	accuracy1 Celsius
Pitch Frame Angle	4	float	[0,360]
Pitch Adjustment Level	4	float	[0,360]

Label	Byte Count	Data Types	Definition
Spare	348	char	String
Servo Mode	1	char	0x11: Pointing mode 0x12: Circular scanning mode 0x27: Pulling out the locking pin 0x88: Inserting a pin
Latitude of the Erection Point	8	double	Latitude of the erection point
Longitude of the Erection Point	8	double	Longitude of the erection point
Elevation of the Erection Point	4	float	Elevation of the installation point
Heading Angle	4	float	The geographical azimuth (heading angle) as defined by the radar array normal.
Valid Heading Mark	1	unsigned char	0: Invalid, 1: Efficient
Number of Satellites	1	unsigned char	The number of satellites detected
North Correction	2	unsigned short	The north correction angle, calculated autonomously by the radar, can be used directly by the user in standby mode. Range: 0–360°
Roll Angle of the Formation	4	float	Corresponding built-in inclinometer
Pitch Angle of the Array	4	float	Corresponding built-in inclinometer

Label	Byte Count	Data Types	Definition
Q/Y/Z	4	float	
HH	2	short	Hour
MM (Month)	2	short	
SS (Seconds)	2	short	Milliseconds (range: 90–99) represent 0.90ms precision (10ms)
M5 (Milliseconds)	2	short	Milliseconds (range: 90–99) represent 0.90ms precision (10ms)
YEAR	2	short	Year
MONTH	2	short	Month
DAY	2	short	Day
Servo Status	1	unsigned char	Bit0 indicates normal, Bit1: Anomaly status
Scanning Mode	1	unsigned char	Bit0 indicates sweep mode, Bit1: Phase sweep, Bit2: Backup sweep
Radar IP Address	16	unsigned char	IP Address in ASCII format
Radar Port Number	2	unsigned short	Radar port number
Client IP Address	16	unsigned char	Client's IP address in ASCII format
Client Port Number	2	unsigned short	Client's port number

Label	Byte Count	Data Types	Definition
Reserve	4	char	

3.2.3 Search target information frame

Each radar pulse group sends target information to the display and control system.

This frame is of variable length, its length depending on the number of targets. Even when no target is found in the current pulse group, a target search header is still sent.

At this time, there is no target content information.

surface5 Search target information header (head)

Label	Byte Count	Data Type	Definition
Search Location	4	float	
Search Overview	4	float	
Search Cycle	4	unsigned int	
Current Page Number	4	unsigned int	
Target Number	2	unsigned short	Report target numberN
Spare	2	unsigned char	0x0000
Total	20		

Table: SurfaceSearch Target Information Main Content (Body)

Label	Byte Count	Data Type	Definition
TargetTrack Deletion Marker	2	unsigned short	0x00: Do not delete flight tracks 0x01: Track deletion

Label	Byte Count	Data Type	Definition
TargetTrack Number	2	unsigned short	0–Maximum number (10000) (Tentative)
TargetLoss Count	2	unsigned short	
Target Reason for Loss	2	unsigned short	
Target Update Time	4	unsigned int	Unit: CPI count
Target Energy	4	float	32-bit binary floating point number
Target Credit Ratio	4	float	32-bit binary floating point number
Target Speed	4	float	32-bit binary floating point number
Target Distance	4	float	32-bit binary floating point number
Target1 Position	4	float	32-bit binary floating point number
Target1 Looking Up and Down	4	float	32-bit binary floating point number
Target1 High	4	float	32-bit binary floating point number
Target1 Track Type	1	unsigned char	0x00: Unknown target 0x11: People 0x12: Car 0x20: Drones 0x24: Bird 0x31: Vessel 0x40: Other objectives 0x41: False target

Label	Byte Count	Data Type	Definition
Target1 Attributes (Search)	1	unsigned char	
Target1 Radar ID	1	unsigned char	
Spare1	3	unsigned char	
Target1 Track Points	2	unsigned short	
Target Longitude	8	double	Reserved decimal places
Target Latitude	8	double	Reserved decimal places
Target1 High	8	double	Reserved decimal places
Target1 Vx	4	float	
Target1 Vy	4	float	
Target1 Vz	4	float	
Target1 Spare	22	unsigned char	
Target1 Envelope Points	1	unsigned char	
Spare2	13	unsigned char	

Label	Byte Count	Data Type	Definition
Target1 UTC Time Years	2	short	Year
Target1 UTC Time Month	1	char	Month
Target1 UTC Time Day	1	char	Day
Target1 UTC Time Hour	1	char	Hour
Target1 UTC Time Division	1	char	Point (second)
Target1 UTC Time in Seconds	1	char	Scope: 0–99, represents 0–990ms precision 10ms
Target1 UTC Time in Milliseconds	1	char	
Target1 Spare2	19	unsigned char	
Target1 Confidence	1	unsigned char	Accuracy: 0.01
TargetN Track Deletion Marker	2	unsigned short	0x00: Do not delete flight tracks 0x01: Track deletion
TargetN Track Number	2	unsigned short	0–Maximum number (10000) (Tentative)

Label	Byte Count	Data Type	Definition
TargetLoss Count	2	unsigned short	
Target Reason for Loss	2	unsigned short	
Target Update Time	4	unsigned int	Unit: CPI count
Target Energy	4	float	32-bit binary floating point number
Target Credit Ratio	4	float	32-bit binary floating point number
Target Speed	4	float	32-bit binary floating point number
Target Distance	4	float	32-bit binary floating point number
Target Position	4	float	32-bit binary floating point number
Target Looking up and Down	4	float	32-bit binary floating point number
Target High	4	float	32-bit binary floating point number
Target Track Type	1	unsigned char	0x00: Unknown target 0x11: People 0x12: Car 0x20: Drones 0x24: Bird 0x31: Vessel 0x40: Other objectives
Target Attributes (Search)	1	unsigned char	

Label	Byte Count	Data Type	Definition
Target Radar ID	1	unsigned char	
Spare1	3	unsigned char	
Target Track Points	2	unsigned short	
Target Longitude	8	double	Reserved decimal places
Target Latitude	8	double	Reserved decimal places
Target High	8	double	Reserved decimal places
Target NVx	4	float	
Target NVy	4	float	
Target NVz	4	float	
Target Spare1	22	unsigned char	
Target Envelope Points	1	unsigned char	
Spare2	13	unsigned char	
Target UTC Time Years	2	short	Year

Label	Byte Count	Data Type	Definition
Target UTC Time Month	1	char	Moon
Target UTC Time Day	1	char	Day
Target UTC Time	1	char	Hour
Target UTC Time Division	1	char	Point (second)
Target UTC Time in Seconds	1	char	Scope: 0–99, represents 0–990ms precision 10ms
Target UTC Time in Milliseconds	1	char	
Target Spare2	19	unsigned char	
Target Confidence	1	unsigned char	Accuracy: 0.01

3.2.4 Tracking target information frame

The radar sends target tracking information to the display and control system. Even when there is no target in the current pulse group, it still sends target head tracking information; at this time, no...

There is a target content information section.

surface7 Tracking target information header (head)

Label	Byte Count	Data Type	Definition
Search Location	4	float	
Search Overview	4	float	
Search Cycle	4	unsigned int	
Current Page Group Number	4	unsigned int	
Target Number	2	unsigned short	Report the target number (mechanical scanning mode is fixed as follows)
Spare	2	unsigned char	0x0000
Total	20		

Table: SurfaceTracking Target Information Main Content (Body)

Label	Byte Count	Data Type	Definition
Track Deletion Marker	2	unsigned short	0x00: Do not delete flight tracks 0x01: Track deletion
Track Number	2	unsigned short	0–Maximum number (10000) (Tentative)
Loss Count	2	unsigned short	
Reason for Loss	2	unsigned short	

Label	Byte Count	Data Type	Definition
Update Time	4	unsigned int	Unit: CPI count
Target Energy	4	float	32-bit binary floating point number
Target Signal to Noise Ratio	4	float	32-bit binary floating point number
Target Speed	4	float	32-bit binary floating point number
Target Distance	4	float	32-bit binary floating point number
Target Location	4	float	32-bit binary floating point number
Target Pitch	4	float	32-bit binary floating point number
Target Height	4	float	32-bit binary floating point number
Target Track Type	1	unsigned char	0x00: Unknown target 0x11: People 0x12: Car 0x20: Drones 0x24: Bird 0x31: Vessel 0x40: Other objectives
Target Attributes (Search)	1	unsigned char	
Target Radar ID	1	unsigned char	
Spare1	3	unsigned char	
Target Track Points	2	unsigned short	
Target Longitude	8	double	Reserved decimal places

Label	Byte Count	Data Type	Definition
Target Latitude	8	double	Reserved decimal places
Target Height	8	double	Reserved decimal places
Target1 Vx	4	float	
Target1 Vy	4	float	
Target1 Vz	4	float	
Target Backup1	22	unsigned char	
Target Envelope Points	1	unsigned char	
Spare2	13	unsigned char	
Target UTC Time Years	2	short	Year
Target UTC Time Month	1	char	Moon
Target UTC Time Day	1	char	Day
Target UTC Time	1	char	Hour
Target UTC Time Division	1	char	Point (second)
Target UTC Time in Seconds	1	char	Scope: 0–99, represents 0–990ms precision 10ms

Label	Byte Count	Data Type	Definition
Target UTC Time in Milliseconds	1	char	
Target Spare2	19	unsigned char	
Target Confidence	1	unsigned char	Accuracy: 0.01
Target UTC Time in Milliseconds	1	char	Scope: 0–99, represents 0–990ms precision 10ms
Target Backup2	20	unsigned char	
Total	N*148		

3.2.5Parameter update frame

When the radar is in search mode, the user can update some data processing thresholds. The radar will update the track based on the latest settings.

Information. This frame can be sent repeatedly.

surface9Parameter update frame

Label	Byte Count	Data Types	Definition
Spare	317	char	

Label	Byte Count	Data Types	Definition
3kmInner Height Lower Limit	2	short	accuracy1m used for rejection 3km within (inclusive) Targets at altitudes lower than this height
3kmExternal Height Lower Limit	2	short	accuracy1m used for rejection 3km Targets at altitudes lower than this height
Height Limit	2	short	accuracy1m removes targets with height not less than this value.
Speed Limit	2	unsigned short	accuracy1m limits targets with absolute speed lower than this value.
Spare	2	unsigned char	
Target Reporting Distance Limit	2	unsigned short	accuracy1m remove targets that are more than this distance.
Target Reporting Distance Lower Limit	2	unsigned short	accuracy1m remove targets that are less than this distance.
Reserve	74	char	
Total	407		

3.2.6 Scan mode switching command frame

The scan mode switching command is sent from the display control to the data processing software. After receiving the command, the data processing software will safely exit the current mode and enter the scan mode specified by the command. If a duplicate command is received, it will not respond.

The data processing software feeds back the current scanning method and status information to the display and control system in real time. The display and control system then displays and makes decisions based on the data.

The servo can only be controlled in mechanical scanning mode. Therefore, before switching to phase scanning mode, the display and control system must point the servo to the user's

expected pointing angle.

In phase scan mode, the current angle of the servo cannot be obtained by default. GPS And inclinometer data.

surface10Scan mode switching command frame

Label	Byte Count	Data Types	Definition
Scanning Mode	1	unsigned char	0x01: Enter mechanical scanning mode 0x02: Enter phase scan mode 0x03: Enter mechanical phase scan mode (backup)
Spare	15	unsigned char	Other values will be treated as mechanical scanning modes.

Appendix 1:

Serial Number	Frequency Code	X Frequency (GHz)	Ku Frequency (GHz)
1	0	9.2	15.70
2	1	9.23	15.75
3	2	9.26	15.80
4	3	9.29	15.85
5	4	9.32	15.90
6	5	9.35	15.95
7	6	9.38	16.00

Serial Number	Frequency Code	X Frequency (GHz)	Ku Frequency (GHz)
8	7	9.41	16.05
9	8	9.44	16.10
10	9	9.47	16.15
11	10	9.5	16.20
12	11	9.53	16.25
13	12	9.56	16.30
14	13	9.59	16.35
15	14	9.62	16.40
16	15	9.65	16.45
17	16	9.68	16.50
18	17	9.71	16.55
19	18	9.74	16.60
20	19	9.77	16.65
21	20	9.8	16.70

Appendix II:

In mechanical scanning standby mode:

Use the current heading angle and the previous30Compare with the mean of the second heading angle; if it is less than...0.01At that time, substitute the current heading angle into the following function to calculate, and...

Calculate the heading angle.

```
float north_angle(float servo_angle, float course_angle) {
```

```
float north = 0.0; float
```

```

angle = 0.0; angle =
course_angle - 90;
while(angle>360)
angle = 360;
while(angle<0)
angle += 360;
north = servo_angle - angle;
while(north>360)
north-=360;
while (north < 0)
north+=360;
return north;
}

```

servo_angle:3.2.2Status information frame radar front end1Azimuth Angle

course_angle:3.2.2State information frame heading angle